

Chapter 1

Introduction to Differential Equation

1.1 Definition and terminology

(Follows from Dennis G. Zill [1])

Introduction: The function $y = e^{0.05x}$, with independent variable x and dependent variable y , is an exponential function in which the function increases with 5% growth rate compounded continuously. The function is differentiable on $(-\infty, \infty)$ and by differentiation

$$\frac{dy}{dx} = 0.05y, \quad (1)$$

$\therefore y = e^{0.05x}$ is a solution to the equation (1). Equation (1) contains derivative $\frac{dy}{dx}$ and it is thus called a *Differential Equation* (DE).

Definition 1.1.1 Differential Equation

An equation containing the derivatives of one or more dependent variables, with respect to one or more independent variables, is said to be a *Differential Equation* (DE).

Classification by Type.

Ordinary Differential Equation (ODE): Contains ordinary derivatives of one or more dependent variable with respect to a single independent variable.

Examples: $\frac{dy}{dx} + 5y = e^x$, $\frac{d^2y}{dx^2} - \frac{dy}{dx} + 6y = 0$, (2)

Partial Differential Equation (PDE): Contains Partial derivatives of one or more dependent variable with respect to two or more independent variable.

Examples: Poisson Equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$ (3)

From now on DE refer only to ODE.

Classification by Order.

Order of DE is the order of the highest derivative in the DE. For example,

$$\frac{d^2y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 6y = e^x$$

is a second order DE.

37 In symbols we express n th-order DE in one dependent
38 variable by the general form

$$39 \quad F(x, y, y', \dots, y^{(n)}) = 0, \quad (4)$$

40 **Classification by Linearity:** An n th-order DE (4) is said to
41 be **linear** if F is linear in $y, y', \dots, y^{(n)}$.

42 That is if (4) is in the form

$$43 \quad a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y \\ 44 \quad = g(x) \quad (6)$$

45 The two properties of linear DE are as follows:

- 46 • The dependent variable y and all its derivatives
47 $y', y'', \dots, y^{(n)}$ are of the first degree, that is the
48 power of each term involving y is 1.
- 49 • The coefficients $a_0, a_1, \dots, a_n$ of $y, y', \dots, y^{(n)}$
50 depends at most on the independent variable x .

51 **Example:** $\frac{dy}{dx} + 5y = e^x$, $\frac{d^2 y}{dx^2} - \frac{dy}{dx} + 6y = 0$ are both
52 linear but of first order and second order respectively.
53 While,

$$54 \quad \frac{d^2 y}{dx^2} + \sin y = 0 \text{ and } \frac{d^2 y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 6y = e^x,$$

55 are both second order non-linear DEs.

56

57 **Example 1 (Verification of a solution)**

58 Verify that the indicated function is a solution of the given
59 differential equation on the interval (∞, ∞) .

60 (a) $\frac{dy}{dx} = xy^{\frac{1}{2}}; \quad y = \frac{1}{16}x^4$

61 (b) $y'' - 2y' + y = 0; \quad y = xe^x$

62 (See Example 1 in **Zill**)

63

64

65 **Example 2 (Verification of a solution)**

66 Verify that

67 $x^2 + y^2 = 25$ is an implicit solution of the DE

68 $\frac{dy}{dx} = -\frac{x}{y}; \quad \text{on open interval } (-5, 5)$

69 (See Example 3 in **Zill**)

70

71 **Exercises 1.1**

72 [**Zill Exercise 1.1**]: 1 – 6, 11 – 13.

73

State the order and determine the linearity of the following DE.

1. $(1 - x)y'' - 4xy' + 5y = \cos x$

2. $x \frac{d^3 y}{dx^3} - \left(\frac{dy}{dx} \right)^4 + y = 0$

3.

Verify that the indicated function is a solution of the DE.

11. $2y' + y = 0; \quad y = e^{-\frac{x}{2}}$

13. $y'' - 6y' + 13y = 0; \quad y = e^{3x} \cos 2x$

1.2 Initial Value Problems (IVPs)

(Follows from Dennis G. Zill [1])

Example 1 (First order IVP)

$y = ce^x$ is a one-parameter family of solutions on $(-\infty, \infty)$ of the simple first order equation $y' = y$. If we impose an initial (side) condition, say, $y(0) = 3$, then $c = 3$. Thus $y = 3e^x$ is a particular solution of the IVP:

$$y' = y; \quad y(0) = 3$$

Now if we impose an initial (side) condition, say, $y(1) = -2$, then $c = -2e^{-1}$. Thus $y = -2e^{x-1}$ is a particular solution of the IVP:

$$y' = y; \quad y(1) = -2$$



Figure 1.2.3

Example 2 (Solution Interval)

In section 2.2, you will learn that the DE $y' + 2xy^2 = 0$ has a one parameter family of solutions

$$y = 1/(x^2 + c) .$$

For i.c. $y(0) = -1$; $y = \frac{1}{x^2 - 1}$ is the soln. of the IVP.

We observe that the function is defined in $\mathbb{R} \setminus \{\pm 1\}$ but the solution interval I of the IVP is $(-1, 1)$.

Figure 1.2.4

Example 3 (Second order IVP)

One can verify that

$$x(t) = c_1 \cos 4t + c_2 \sin 4t$$

is a two parameter family of solutions of

$$x'' + 16x = 0.$$

To obtain a particular solution, one needs to impose two conditions, say,

$$(i) \quad x\left(\frac{\pi}{2}\right) = -2, \quad (ii) \quad x'\left(\frac{\pi}{2}\right) = 1.$$

Using (i) and (ii) in (1), $c_1 = -2$; $c_2 = 1/4$.

Hence $x(t) = -2 \cos 4t + \frac{1}{4} \sin 4t$ is particular solution of (2) with initial conditions (i) and (ii).

Exercise 1.2

[Zill Exercise 1.2]: 3–4, 7–8.

$y = 1/(x^2 + c)$ is a one parameter family of solutions.

$y' + 2xy^2 = 0$. Find the solution and the largest solution

interval of the IVP with

$$(3) y(2) = 1/3 \quad (4) y(-2) = 1/2$$

Verify that $x(t) = c_1 \cos t + c_2 \sin t$ is a two-parameter

family of solutions of $x'' + x = 0$. Find a solution of the

second order IVP with

$$(7) x(0) = -1, x'(0) = 8; \quad (8) x(\pi/2) = 0, x'(\pi/2) = 1$$

1.3 Differential Equations as Mathematical Models

(Follows from Dennis G. Zill [1], Braun [2])

Mathematical models describe the behavior of some real-life system or phenomenon (physical, sociological, economic) in mathematical terms. Construction of mathematical model of a system starts with

- (i) Identification of the variables and parameters that are responsible for changing the system. We may choose not to incorporate all these variables into the model at first.
- (ii) Next, we make a set of reasonable assumptions, or hypotheses about the system.

POPULATION DYNAMICS

Variables are time t , population size $P(t) :=$ the total population at time t , is a function of time t . Parameter, $a :=$ growth rate

1.3.1 Malthusian Population Model

Malthusian population model is based on the assumption that the rate at which the population grows is proportional to $P(t)$. Mathematically,

$$\frac{dP}{dt} \propto P$$

$$\text{Or} \quad \frac{dP}{dt} = aP, \quad (3.1)$$

Where the constant of proportionality $a = \frac{1}{p} \frac{dP}{dt}$ is a parameter in the model which denotes (specific) growth rate of the population.

This simple model fails to take into account immigration, emigration, carrying capacity, etc., nevertheless, it is still used to model growth of small populations over short intervals of time.

(3.1) is a *first order linear* ODE.

Initial value problem

ODE (3.1) appended with initial condition and reads as an initial value problem (IVP) as follows.

If the population of a given species is p_0 at time t_0 , then $p(t)$ satisfies the *initial value problem (IVP)*:

$$\frac{dP}{dt} = aP; \quad P(t_0) = P_0 \quad (3.2)$$

Solution of Malthusian Model:

$$P(t) = p_0 e^{a(t-t_0)} \quad (3.3)$$

Hence any species satisfying the **Malthusian Model** of population growth grows exponentially with time t .

Application of the exponential model

- Microbiology (growth of bacteria),
- Conservation biology (restoration of distributed populations),
- Fishery (prediction of fish dynamics).

The limitation with Malthus's Model

Malthus's population model predicts either population growth without bound or inevitable extinction. The difference is based on whether the growth rate a is positive or negative. Neither case is typically observed in genuine biological communities. What is often observed instead is that small populations often (though not always) increase

218 in number while very large populations tend to decline in
219 number. In both cases a steady state is often reached after
220 which significant changes in population are not observed
221 unless a significant environmental change is encountered.
222 A good population model must therefore reproduce this
223 behavior.

224 From a biological point of view the missing feature of
225 Malthus's model is the idea of **carrying capacity**. As a
226 population increases in size the environment's ability to
227 support the population decreases. As the population
228 increases per capita food availability decreases, waste
229 products may accumulate and birth rates tend to decline
230 while death rate tend to increase. It seems reasonable to
231 consider a mathematical model which explicitly
232 incorporates the idea of carrying capacity.

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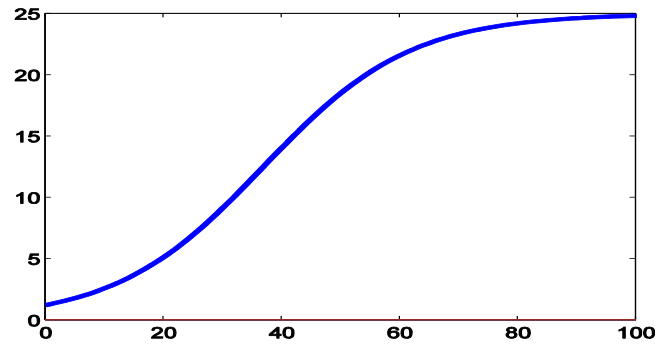
1.3.2 The Logistic Population Model:

In this section we discuss in detail a model for population growth, the logistic model, which is more sophisticated than exponential growth. The logistic model, a slight modification of Malthus's model, is just such a model. When the population gets extremely large though, Malthus's model cannot be very accurate, since they do not reflect the fact that individual members are now competing with each other for the limited living space, natural resources food available. Thus, we must add a competition term to our linear differential equation. A suitable choice of a competition term is $-bP^2$, where b is a constant, since the statistical average of the number of encounters of two members per unit time is proportional to P^2 . We consider, therefore the modified equation

$$\frac{dP}{dt} = aP - bP^2 \quad \dots \quad \dots \quad \dots \quad (3.4)$$

252 This equation is known as the logistic law of population
 253 growth and the numbers **a** , **b** are called the vital coefficients
 254 of the population. It was first introduced in 1837 by the
 255 Dutch mathematical-biologist Verhulst. Now, the constant
 256 **b** , in general, will be very small compared to **a** , so that if p
 257 is not too large then the term $-bP^2$ will be negligible
 258 compared to ap and the population will grow exponentially.
 259 However, when p is very large, the term $-bP^2$ is no longer
 260 negligible, and thus serves to slow down the rapid rate of
 261 increase of the population. Needless to say, the more
 262 industrialized a nation is, the more living space it has, and
 263 the more food it has, the smaller the coefficient b is. The
 264 constant can be expressed in terms of a as: $b = \frac{a}{k}$, where k
 265 is **carrying capacity** or upper limit of population. The
 266 logistic model (1.4) is a ***first order nonlinear*** ODE.
 267 If P_0 is the population at time t_0 , then $p(t)$, the population
 268 at time , satisfies the initial-value problem

$$\begin{aligned} \frac{dP}{dt} &= ap - bP^2 \quad \dots \quad \dots \quad \dots \quad (3.5) \\ P(t_0) &= P_0 \end{aligned}$$



1.3.3 RADIOACTIVE DECAY

$A(t)$: = Amount of radioactive substance at time t .

$$\frac{dA}{dt} \propto A$$

$$\text{Or} \quad \frac{dA}{dt} = aA, \quad (3.6)$$

Equation (3.1) and (3.6) are same but for growth in (3.1)

$a > 0$ and for decay, in (1.6) $a < 0$.

1.3.4 NEWTON'S LAW OF COOLING/WARMING

Temperature of a body changes is proportional to the difference between the temperature of the body and the temperature of the surrounding medium.

$$\frac{dT}{dt} = a(T - T_m) \quad (3.7)$$

If T_m is a constant,

(i) $T > T_m \Rightarrow \text{cooling} \Rightarrow \frac{dT}{dt} < 0 \Rightarrow a < 0$

(ii) $T < T_m \Rightarrow \text{warming} \Rightarrow \frac{dT}{dt} > 0 \Rightarrow a < 0$

(3.7) is also linear.

1.3.5 SPREAD OF DISEASE

Let $x(t) :=$ Number of people who have contracted the disease and

$y(t) :=$ Number of people who have not yet been exposed.

$$\frac{dx}{dt} = axy \quad (3.8)$$

If $n :=$ total number of population, then $n = x + y$

$\Rightarrow y = n - x$, then (1.8) reads as

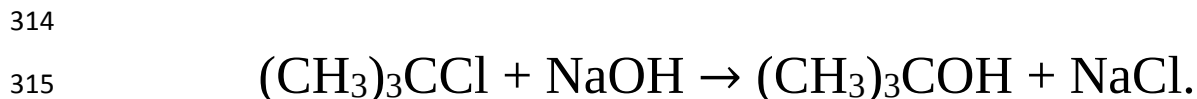
$$\frac{dx}{dt} = ax(n - x) \quad (3.9)$$

This model is also used for the prediction of the spread of technological innovations. (3.8) or (3.9) is non-linear.

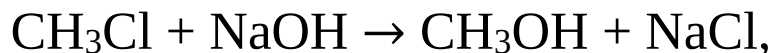
1.3.6 CHEMICAL REACTIONS

The disintegration of radioactive substance, governed by the differential equation (1.6), is said to be a **first-order reaction**. In chemistry a few reactions follow this same empirical law: If $x(t)$ denotes the amount of substance A remaining at any time t , then $\frac{dx}{dt} = kx$, $k < 0$ is a constant.

An example of a first order chemical reaction is the conversion of t -butyl chloride, $(\text{CH}_3)_3\text{CCl}$, into t -butyl alcohol, $(\text{CH}_3)_3\text{COH}$:



Only the concentration of the t -butyl chloride controls the rate of reaction. But in the reaction



one molecule of sodium hydroxide, NaOH is consumed for every molecules of methyl chloride, CH₃Cl, thus forming *one* molecule of methyl alcohol, CH₃OH, and one molecule of sodium chloride, NaCl. In this case the rate at which the reaction proceeds is proportional to the product of the remaining concentrations of CH₃Cl and NaOH.

To describe the second reaction **in general**, let us suppose *one* molecule of a substance *A* combines with *one* molecule of a substance *B* to form *one* molecule of substance *C*. If $x(t)$ denotes the amount of the chemical *C* formed at time *t* and if *a* and *b* are the amounts of *A* and *B* at *t* = 0, then the instantaneous amounts of *A* and *B* not converted to *C* are *a* − *x* and *b* − *x*, respectively. Hence the rate of formation of *C* is given by

$$\frac{dx}{dt} = k(a - x)(b - x) \quad (3.10)$$

where *k* is a constant of proportionality. A reaction governed by the model (1.10) is said to be **second order reaction** and the model (1.10) is a non-linear first order DE.

1.3.7 MIXTURES

The mixing of two salt solutions of differing concentrations gives rise to a first-order differential equation for the amount of salt contained in the mixture. Let us suppose that a large mixing tank initially holds 300 gallons of brine (that is, water in which a certain number of pounds of salt has been dissolved). Another brine solution is pumped into the large tank at a rate of 3 gallons per minute; the concentration of the salt in this inflow is 2 pounds per gallon. When the solution in the tank is well stirred, it is pumped out at the same rate as the entering solution. See Figure 1.3.1. If $A(t)$ denotes the amount of salt (measured in pounds) in the tank at time t , then the rate at which $A(t)$ changes is a net rate:

Figure: 1.3.2 Mixing Tank

$$\begin{aligned}\frac{dA}{dt} &= (\text{Input rate of salt}) - (\text{Output rate of salt}) \\ &= R_{in} - R_{out}\end{aligned}\tag{3.11}$$

Where,

$$R_{in} = (2 \text{ lb/gal}) \cdot (3 \text{ gal/min}) = (6 \text{ lb/min})$$

$$R_{out} = \left(\frac{A(t)}{300} \text{ lb/gal} \right) \cdot (3 \text{ gal/min}) = \frac{A(t)}{100} \text{ lb/min}$$

373

374 Then (3.11) becomes

375

$$\frac{dA}{dt} = 6 - \frac{A}{100}$$

377

$$\text{or} \quad \frac{dA}{dt} + \frac{A}{100} = 6 \quad (3.12)$$

379

380 (3.12) is a first order inhomogeneous DE.

381

1.3.8 SERIES CIRCUITS

Consider the single-loop series circuit shown in Figure (a) containing an inductor, resistor, and capacitor. The current in a circuit after a switch is closed is denoted by $i(t)$; the charge on a capacitor at time t is denoted by $q(t)$. The letters L , R , and C are known as inductance, resistance, and capacitance, respectively, and are generally constants. Now according to **Kirchhoff's second law**, the impressed voltage $E(t)$ on a closed loop must equal the sum of the voltage drops in the loop. Figure (b) shows the symbols and the formulas for the respective voltage drops across an inductor, a capacitor, and a resistor. Since current $i(t)$ is related to charge $q(t)$ on the capacitor by $i = dq/dt$, adding the three voltages

Figure 1.3.4

$$\begin{array}{lll} \text{Inductor: } L \frac{di}{dt} = L \frac{d^2q}{dt^2}, & \text{Resistor: } iR = R \frac{dq}{dt}, & \text{and} \\ \text{Capacitor: } \frac{1}{C} q \end{array}$$

and equating the sum to the impressed voltage yields a second-order differential equation

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t) \quad (3.13)$$

1.3.9 FALLING BODIES

Let $s(t)$ denotes the position relative to the ground at time t .

Newton's second law gives

$$m \frac{d^2 s}{dt^2} = -mg \quad \text{or} \quad \frac{d^2 s}{dt^2} = -g \quad (3.14)$$

$$\text{(IVP)} \quad \frac{d^2 s}{dt^2} = -g, \quad s(0) = s_0, \quad s' = v_0 \quad (3.15)$$

$$\Rightarrow s(t) = -\frac{1}{2}gt^2 + v_0 t + s_0 \quad (3.16)$$

With air resistance: $m \frac{d^2 s}{dt^2} = mg - k \frac{ds}{dt}$

$$\text{or} \quad m \frac{d^2 s}{dt^2} + k \frac{ds}{dt} = mg \quad (3.17)$$

(3.15), (3.17) are second order DEs.

428

429 **1.3.10 HARMONIC MOTION**

430 In this section we discuss simple harmonic motion and
431 damped harmonic motion solution of their differential
432 equations and qualitative behavior of the solution.

433 **Simple harmonic motion**

434 **Figure 5.1.1 Figure 5.1.2**

435

436 Suppose that a flexible spring is suspended vertically
437 from a rigid support and then a mass m is attached to its
438 free end .the amount of stretch or elongation of the spring
439 will of course depend on mass. By Hook's law the spring
440 itself exerts a restoring force opposite to the direction of
441 the elongation and proportional to the amount of
442 elongation, thus

443
$$F = -k(s + x) + mg = -kx$$

where k is a constant called spring constant
and $mg - ks = 0$ is the condition of equilibrium

Now from Newton's 2nd law we know that

$$F = mf$$

Then
$$m \cdot \frac{d^2x}{dt^2} = -kx$$

$$\Rightarrow \frac{d^2x}{dt^2} + \omega^2 x = 0 \text{ where } \omega^2 = \frac{k}{m}.$$

Damped harmonic motion

In mechanics damping forces acting on a body are considered to be proportional to a power of the instantaneous velocity. In particular, we shall assume that this force is given by a constant multiple of $\frac{dx}{dt}$. When no other external force are impressed on the system, it follows from Newton's 2nd law that,

$$m \frac{d^2x}{dt^2} = -kx - \beta \frac{dx}{dt}$$

Where β is the damping constant and negative sign is a consequence of the fact that the damping force acts in a direction to the motion

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x + \frac{\beta}{m}\frac{dx}{dt} = 0$$

$$\Rightarrow \frac{d^2x}{dt^2} + 2\lambda \frac{dx}{dt} + \omega^2 x = 0, \text{ where } 2\lambda = \frac{\beta}{m}$$

$$\text{and } \omega^2 = \frac{k}{m}$$

Exercise 1.3 [Zill]: 1 – 4, 7 – 8, 15 – 16.

References

- 1.1. A First Course in Differential Equations with Modeling and Applications, (10th Edition), Author-Dennis G. Zill.
2. Differential Equations and their Applications, M. Braun, Springer.